A Multilevel Encoder–Decoder Attention Network for Change Detection in Hyperspectral Images

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***Abstract*— Convolutional neural networks (CNNs) have attracted much attention in change detection (CD) for their superior feature learning ability. However, most of the existing CNN-based CD methods adopt an early- or late-fusion strategy to fuse low-level spatial details or high-level semantic information. So far, the impact of multilevel fusion strategy across multitemporal hyperspectral (HS) images, and its application to CD, remains unexplored. In this article, we propose a multilevel encoder–decoder attention network (ML-EDAN), which allows the network to make full use of the hierarchical features for CD in HS images. A two-stream encoder–decoder framework is taken as the backbone to exploit and fuse the hierarchical features from all the convolutional layers of multitemporal HS images. Within the encoder–decoder, a contextual-information-guided attention module is developed to yield more effective spatial–spectral feature transfer in the network. After fully obtaining the multilevel hierarchical features, the long short-term memory (LSTM) subnetwork is devised to analyze temporal dependence between multitemporal images. Moreover, the proposed ML-EDAN is trained in an end-to-end manner with a new joint loss function considering both reconstruction error and pixelwise classification error. The experiments are conducted on three datasets, demonstrating the effectiveness of the proposed ML-EDAN in HS CD in comparison with widely accepted state-of-the-art methods.**

***Index Terms*— Attention mechanism, change detection (CD), deep learning (DL), hyperspectral image (HSI), multilevel features.**

I. Introduction

WITH the rapid development of the remote sensing technology, the remote sensing imagery is more beneficial for us to get even more abundant information about ground objects [1]. It is important to analyze this information accurately for understanding the profound impact of rapid population growth and urbanization on the resources and environment. Change detection (CD) is an effective tool to observe the changes of the Earth's surface, as it could identify differences of regional surface by comparing a set of images obtained at different times for the same scene. Hyperspectral image (HSI) contains rich spectral information, which can be used to distinguish ground objects with similar spectra, so it becomes the main data source for CD. Therefore, HSI-CD has become one of the most active research subjects in remote sensing image processing and has been widely used in many application fields, such as land cover transitions [2], environment monitoring [3], and natural disasters assessment [4].

In recent years, a variety of methods have been proposed to obtain satisfactory results of HSI-CD. The traditional approaches can be classified into the following four categories: 1) image algebra; 2) transformation-based methods; 3) classification; and 4) other advanced algorithms. The image algebra methods, which measure the intensity of ground-object changes by directly performing mathematical operations on the corresponding bands, are very fast and easy to implement. They use techniques such as image subtraction, image regression, and change vector analysis (CVA) [5] to generate different images (DIs). However, the image algebra methods are susceptible to radiation and noise, and it is difficult to select an appropriate threshold [6]. The transformation-based methods are proposed to extract effective features and reduce data redundancy by transforming HSIs to other feature spaces, where the changes can be enlarged. Principal component analysis (PCA) [6], slow feature analysis (SFA) [7], and iteratively reweighed multivariate alteration detection (IR-MAD) [8] are the most popular ones among the transformation-based methods. PCA is a widely used subspace learning algorithm, which selects the most representative features to retain key information and reduce noise interference. Nevertheless, the generation of new components will be time-consuming if the data are unbalanced. Classification methods contain a compound classification approach [9] and postclassification comparison [10], [11], which requires prior knowledge for training the classifier and provides complete "from-to" information. Postclassification is a supervised method, which uses the trained classifier to classify the bitemporal images independently and compares the generated class maps pixel by pixel to get the changed regions. Although the classification-based methods are less sensitive to noise,

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the performance of these approaches depends heavily on the accuracy of classification algorithm [12]. In addition, CD can also be implemented by some advanced models, such as conditional random filed [13], [14], tensor factorization [15], [16], and low-rank and sparse representation [17]. Although these traditional methods are constantly updated, they are difficult to identify the critical information in the high-dimensional data that represent the change of ground features.

Deep learning (DL) has achieved great success in various fields, such as computer vision [18] and remote sensing tasks [19]–[21]. Compared with the traditional CD models, DL models are capable of mining abstract features hierarchically from different layers and solving the problem of high dimension. Convolutional neural network (CNN) is a classical DL architecture, which allows the network to extract spatial–spectral features from input in an end-to-end manner. An optical aerial image CD method proposed by Zhan *et al.* [22] makes use of a deep Siamese convolutional network, in which a weighted contrastive loss is utilized to balance the numbers of changed and unchanged samples. Wang *et al.* [23] introduced a mixed-affinity matrix as the input of the network, to learn representative cross-channel features from multisource data. The recurrent neural network (RNN), which offers the potential for processing sequential time series data, is also introduced to CD tasks for further mining the temporal connection. Lyu *et al.* [24] proposed an RNN-based network, called REFEREE, which learns a reliable change rule to represent the difference information and provides the transfer capacity to detect changes in new target images. Moreover, Mou [25] proposed a recurrent CNN (ReCNN), which first combines the CNN and RNN together, to detect changes by learning a spectral–spatial–temporal feature representation. Song *et al.* [26] proposed a CD method called recurrent 3-D fully convolutional network (Re3FCN), which combines the superiorities of the 3-D convolutional network and the convolutional long short-term memory (LSTM). Lin *et al.* [27] proposed the bilinear convolutional neural networks (BCNNs) to detect changes in bitemporal multispectral images. BCNNs perform matrix cross multiplication on the output feature maps of the two images to obtain the combined bilinear features and then inputs the combined features into the classifier to obtain the final results.

Although the existing DL methods have made outstanding contributions to the development of CD, there still exists one severe issue to address. The features of HSIs cannot fully be extracted by adopting these methods. The existing DL methods generally follow an early- or late-fusion strategy, which mainly only focuses on fusing the low-level spatial information or the high-level semantic information. Since the HSIs contain complex spatial information and ground details, both low-level features with fine details and high-level features containing semantic information should be taken into consideration. It is necessary to explore a multilevel fusion strategy to make full use of the features from different layers for the HSI CD tasks.

Aiming at the above problem, this article develops a new CD network, namely, multilevel encoder–decoder attention network (ML-EDAN), which has the following main contributions.

1) We propose an ML-EDAN for HSI-CD, which allows the network to make full use of the hierarchical low- and high-level features from the multitemporal HSIs to achieve the accurate recognition.
2) Two-stream encoder–decoder subnetwork is designed, and the contextual-information-guided attention (CIGA) module is embedded in the encoder–decoder subnetwork to effectively exploit and fuse the multilevel spatial–spectral features.
3) The proposed ML-EDAN method makes the combination of the designed encoder–decoder subnetwork with LSTM subnetwork to extract the multilevel temporal–spatial–spectral features, which not only yields the spatial–spectral characteristics of the two HSIs but also considers the temporal correlation between bitemporal images.
4) A new loss function is defined to integrate reconstruction error and pixelwise classification error together so that the network can fully learn effective features for accurate label prediction.

The remainder of this article is organized as follows. Section II is dedicated to the details of the proposed method. In Section III, the performance of the proposed method is evaluated on three datasets. Section IV draws the conclusion of this article.

II. Proposed Method

In this section, we propose an ML-EDAN for HSI CD task, which has a strong ability in spatial–spectral–temporal feature learning. The proposed ML-EDAN can be directly trained in an end-to-end manner without generating a difference map. The overall architecture, as shown in Fig. 1, is composed of a multilevel encoder–decoder subnetwork with CIGA module and an LSTM subnetwork. Two parallel encoder–decoder architectures with CIGA modules are employed to extract multilevel hierarchical features from the bitemporal images. The extracted multilevel features are fed into the LSTM subnetwork to extract and fuse the temporal features of the two images adaptively and jointly. Furthermore, a new loss function is proposed to better learn feature representations. The architecture and the loss function of ML-EDAN will be described in detail in the following.

A. Multilevel Spatial–Spectral Feature Extraction via Encoder–Decoder Subnetwork

1) Architecture of the Multilevel Encoder–Decoder Subnetwork: Because of its powerful data modeling and feature learning capabilities, CNN is widely used in HSI CD tasks to extract the spatial–spectral features of HSIs. However, most of the existing CNN-based methods integrate single-level features at the input or the output of the network, which leads to the loss of useful information. Low-level features mainly characterize spatial details [28], [29], whereas high-level features refine semantic information, which is too coarse to generate fine boundaries.

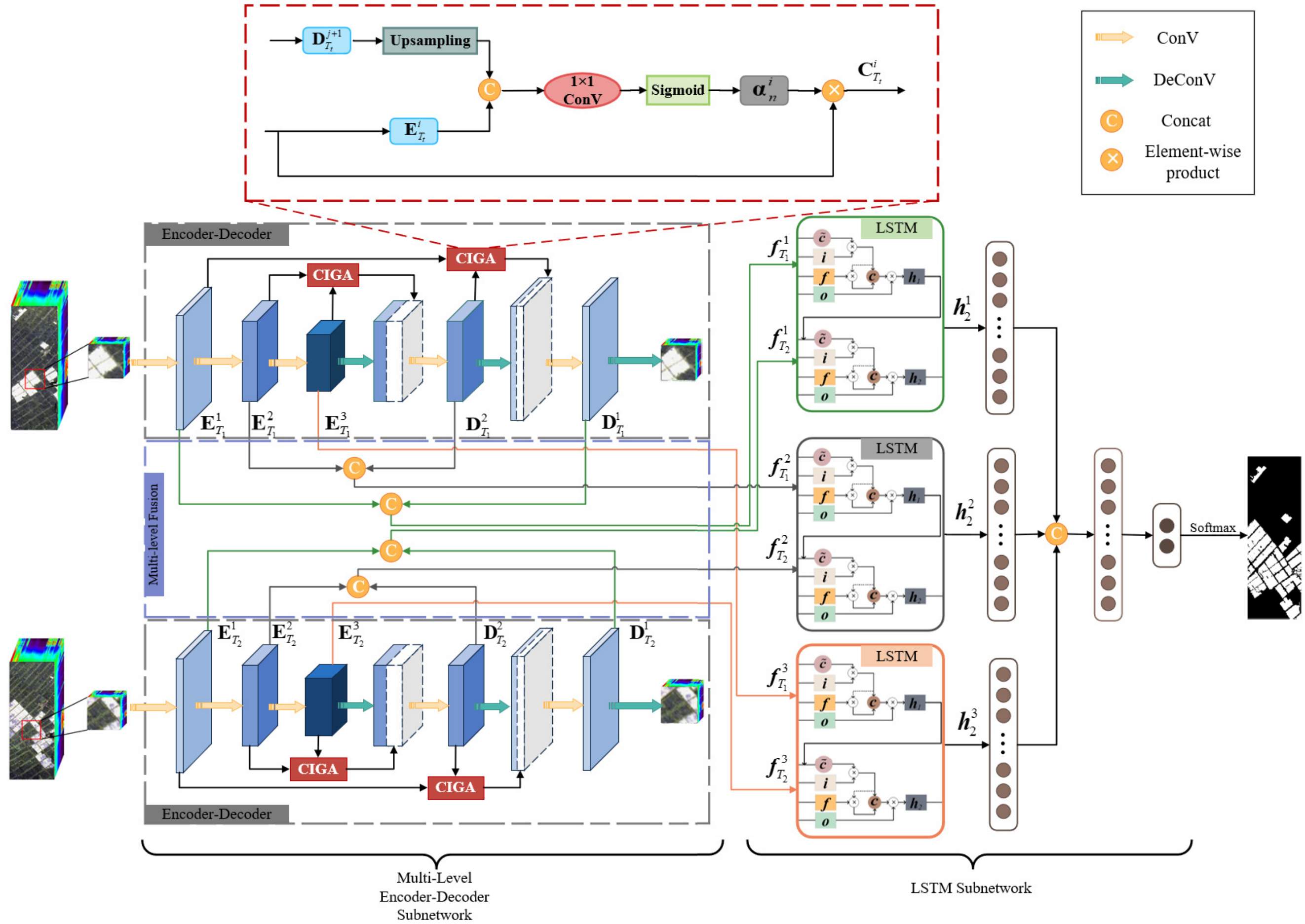

Fig. 1. Architecture of the proposed multilevel recurrent encoder–decoder attention networks, which consists of a multilevel encoder–decoder subnetwork and an LSTM-based subnetwork.

Motivated by these observations, we propose a novel two-stream encoder–decoder framework to exploit and fuse the hierarchical features from all the convolutional layers, so as to provide more abundant multilevel spatial-spectral information. The encoder with multiple convolutional layers learns the effective features representation of the input data, whereas the decoder utilizes the deconvolution operation to generate the output that best characterizes the features. To help the decoder better repair the details of the target, the feature maps extracted by encoder are further merged through skip connections level by level. The CIGA module is embedded in the encoder–decoder subnetwork, and CIGA is a variant of attention that aims to extract and transfer the advantageous spectral–spatial information between low level and high level, yielding more efficient spatial–spectral information blending across encoder and decoder. Different from the traditional attentions, the attention maps in CIGA are computed from high- and low-level features since they can provide with more faithful spatial–spectral guidance.

As shown in Fig. 1, the encoder contains a series of convolutional layers with a kernel size of 3 × 3, each followed by a rectified linear unit (ReLU). Given an input patch $\mathbf{I}_{T_t}$, the output feature map of the ith level in encoder can be defined as

$$\mathbf{E}_{T_t}^{i} = f\left(\mathbf{E}_{T_t}^{i-1} * \mathbf{W}_{T_t}^{i} + \boldsymbol{b}_{T_t}^{i}\right), \quad i = 1, 2, 3 \tag{1}$$

where $\mathbf{E}_{T_t}^{i}$ represents the feature map generated by the ith convolutional layer for the input patch $\mathbf{I}_{T_t}$ $(t = 1, 2)$, $*$ denotes the convolutional operation, $\mathbf{W}_{T_t}^{i}$ and $\boldsymbol{b}_{T_t}^{i}$ are the learned weight and bias of the ith convolution layer on the input patch $\mathbf{I}_{T_t}$, respectively, and $f(\cdot)$ denotes the active function (we use ReLU in this work), which can be described as

$$f(x) = \max(0, x) \tag{2}$$

where x represents the value of feature map.

The decoder maps the latent representation to a high-dimensional space to reconstruct the input patches of the encoder. It is composed of 3 × 3 convolutional layers for feature fusion and 3 × 3 deconvolutional layers for feature reconstruction. Instead of simply concatenating the feature maps of encoder and decoder on the same spatial scale, we further utilize the CIGA modules to yield more effective spatial–spectral feature transfer in the encoder–decoder network. The skip connections with embedded attention are able to bring pixel-level positioning preserved in the early encoder layer, which helps to better recover detail structures of the inputs. The output feature map of the jth level in decoder can be formulated as the following process:

$$\widetilde{\mathbf{D}}_{T_t}^{j} = f\left(\mathbf{D}_{T_t}^{j} * \widetilde{\mathbf{W}}_{T_t}^{j} + \widetilde{\boldsymbol{b}}_{T_t}^{j}\right), \quad j = 3, 2, 1 \tag{3}$$

where $\widetilde{\mathbf{W}}_{T_t}^{j}$ and $\widetilde{\boldsymbol{b}}_{T_t}^{j}$ are, respectively, the weight matrix and bias of the jth deconvolutional layer, $\widetilde{\mathbf{D}}_{T_t}^{j}$ is the output of the jth deconvolutional layer, and $\mathbf{D}_{T_t}^{j}$ represents the jth feature map of the convolutional layer in decoder, which can be expressed as

$$\mathbf{D}_{T_t}^{j} = f\left(\mathrm{Concat}\left(\widetilde{\mathbf{D}}_{T_t}^{j+1}, \mathbf{C}_{T_t}^{i}\right) * \mathbf{W}_{T_t}^{j} + \boldsymbol{b}_{T_t}^{j}\right), \quad i = j = 2, 1 \tag{4}$$

where $\mathbf{W}_{T_t}^{j}$ and $\boldsymbol{b}_{T_t}^{j}$ are the learned weight and bias of the ith convolution layer in decoder, respectively, $\widetilde{\mathbf{D}}_{T_t}^{j+1}$ represents the feature map provided by the $(j+1)$th convolutional layer in decoder, $\mathrm{Concat}(\cdot)$ denotes the concatenation operation, and $\mathbf{C}_{T_t}^{i}$ is the output of the CIGA module.

The encoder extracts the feature from the input and gradually reduces the spatial dimension, whereas the decoder is responsible for recovering abstract features. In order to make full use of the complementary information, the feature maps of encoder and decoder at the same spatial resolution are concatenated, which can be expressed as

$$\mathbf{F}_{T_t}^{i} = \begin{cases} \mathrm{Concat}\left(\mathbf{E}_{T_t}^{i}, \mathbf{D}_{T_t}^{j}\right), & i = j = 1, 2 \\ \mathbf{E}_{T_t}^{i}, & i = 3 \end{cases} \tag{5}$$

where $\mathbf{F}_{T_t}^{1}$, $\mathbf{F}_{T_t}^{2}$, and $\mathbf{F}_{T_t}^{3}$ denote different-level features of size 5×5, 3×3, and 1×1 that are generated by the encoder–decoder architecture, respectively. Rather than considering the latent representation in a single level, the multilevel features are further employed to model temporal dependence between data of T_1 and T_2. The three pairs of multilevel spatial–spectral features make full use of the information with different resolution so that both high-level semantic information and spatial details with high resolution are taken into consideration.

2) CIGA Module: The attention mechanism is similar to human visual attention mechanism, which automatically focuses on the target area of interest to obtain more detailed information. Since the attention module helps to assign greater weights to features critical for a specific task, the network has the ability to highlight the relevant portions of the image and suppress unrelated information. In recent years, DL network combined with attention modules is widely applied [30], [31], and has been proven to effectively improve the performance of the network.

We propose a novel CIGA module that is designed for the CD task. Contextual information extracted at a deeper level serves as the guidance of the attention module and helps to identify the information that is critical to the task. The architecture of the proposed attention module is shown in Fig. 1. The CIGA is used to filter the feature propagated by the skip connection, which is accomplished through the use of guiding map $\mathbf{D}_{T_t}^{j+1}$ obtained in the decoder. The output of the ith convolutional layer $\mathbf{E}_{T_t}^{i}$ in the encoder is then concatenated to the upsampled guiding map, and the nearest neighbor interpolation is employed to obtain inputs at the same spatial resolution. A channelwise convolutional layer with the size of 1×1 is utilized to perform linear transformation and a sigmoid activation is used to normalize the attention coefficient $\boldsymbol{\alpha}_{n}^{i}$ of each pixel n. The formulation of the attention coefficients is as follows:

$$\boldsymbol{\alpha}_{n}^{i} = \begin{cases} \sigma\left(\mathrm{Concat}\left(\mathbf{E}_{T_t}^{i}, \mathbf{D}_{T_t}^{j+1} \uparrow\right) * \mathbf{W}_{\varphi}^{i} + \boldsymbol{b}_{\varphi}^{i}\right), & i = j = 1 \\ \sigma\left(\mathrm{Concat}\left(\mathbf{E}_{T_t}^{i}, \mathbf{E}_{T_t}^{i+1} \uparrow\right) * \mathbf{W}_{\varphi}^{i} + \boldsymbol{b}_{\varphi}^{i}\right), & i = 2 \end{cases} \tag{6}$$

where $\mathbf{W}_{\varphi}^{i}$ and $\boldsymbol{b}_{\varphi}^{i}$ are the linear transformation and bias of the convolutional layer, respectively, $\uparrow$ denotes the upsampling operation, and $\sigma(\cdot)$ is the sigmoid function, which can be expressed as

$$\sigma(x) = \frac{1}{1 + e^{-x}}. \tag{7}$$

The attention coefficient is capable of modulating feature responses to reserve the activations related to the target task and suppresses the unrelated information. Thus, the output of CIGA $\mathbf{C}_{T_t}^{i}$ can be computed by the elementwise multiplication of the attention coefficient and the input feature map $\mathbf{E}_{T_t}^{i}$

$$\mathbf{C}_{T_t}^{i} = \mathbf{E}_{T_t}^{i} \otimes \boldsymbol{\alpha}_{n}^{i}, \quad i = 2, 1 \tag{8}$$

where $\otimes$ denotes the elementwise product.

B. Multilevel Temporal–Spatial–Spectral Feature Extraction via LSTM Subnetwork

The CNN-based CD methods have achieved impressive success in recent years, while all the inputs are considered to be independent in these networks. The CNN-based methods generally extract the spatial–spectral features of the two images separately but ignore the temporal correlation between bitemporal images. One important issue in CD is to model the temporal correlation between bitemporal images [25]. RNN is capable of extracting temporal features of the bitemporal images by using the recurrent hidden state of the previous time. LSTM, as a special RNN structure, can effectively overcome the problems of gradient disappearance and gradient explosion during the RNN training process [32]–[34]. Therefore, the LSTM-based subnetwork is adopted to extract effective multilevel temporal–spatial–spectral features of bitemporal images, so as to realize the accurate recognition of changed and unchanged regions.

Since the input of the LSTM-based subnetwork is required to be one dimension, the multilevel feature maps $\mathbf{F}_{T_t}^{i}$ are transformed into feature vectors $\boldsymbol{f}_{T_t}^{i}$. As shown in Fig. 1, given a sequence $(\boldsymbol{f}_{T_1}^{i}, \boldsymbol{f}_{T_2}^{i})$, the LSTM is able to remain the effective feature and discards the redundant information. After the multilevel features are fed into the LSTM subnetwork respectively, three sequences with temporal–spatial–spectral features can be obtained and only the last hidden state $\boldsymbol{h}_{2}^{i}$ of each sequence will be reserved.

An LSTM unit is composed of forget gate $\boldsymbol{f}_t$, input gate $\boldsymbol{i}_t$, output gate $\boldsymbol{o}_t$, and memory cell $\boldsymbol{c}_t$. The forget gate $\boldsymbol{f}_t$ allows the unit to forget useless information from the memory cell. At time step t, the forget gate can be described as

$$\boldsymbol{f}_t = \sigma\left(\mathbf{W}_{\mathrm{fi}} \boldsymbol{f}_{T_t}^{i} + \mathbf{W}_{\mathrm{fh}} \boldsymbol{h}_{t-1} + \boldsymbol{b}_f\right) \tag{9}$$

where $\mathbf{W}$ denotes the coefficient matrices with the subscripts having specific meaning. For example, $\mathbf{W}_{\mathrm{fi}}$ is the weight matrix between input $f^i_{T_t}$ and forget gate f_t, $\mathbf{W}_{\mathrm{fh}}$ is the weight matrix between hidden state h_{t-1} and forget gate f_t. $\sigma(\cdot)$ represents the sigmoid function, which makes the outputs vary from zero to one, and $\mathbf{b}_f$ is the bias vector of the forget gate.

The input gate i_t is responsible for determining whether the new information should be added to the memory cell. The input gates are updated by

$$i_t = \sigma\left(\mathbf{W}_{ii} f^i_{T_t} + \mathbf{W}_{ih} h_{t-1} + b_i\right) \tag{10}$$

where $\mathbf{W}_{ii}$ and $\mathbf{W}_{ih}$ are the coefficient matrices of the input gate and b_i is the bias vector.

The memory cell c_t is updated by selectively discarding the previous information memorized in cell state c_{t-1} and adding new information $\tilde{c}_t$

$$c_t = i_t \odot \tilde{c}_t + f_t \odot c_{t-1} \tag{11}$$

where $\odot$ denotes the pointwise multiplication and $\tilde{c}_t$ is the new candidate cell value produced by each cell, which can be represented by

$$\tilde{c}_t = \tanh\left(\mathbf{W}_{\mathrm{ci}} f^i_{T_t} + \mathbf{W}_{\mathrm{ch}} h_{t-1}\right) \tag{12}$$

where $\tanh(\cdot)$ denotes the hyperbolic tangent function, $\mathbf{W}_{\mathrm{ci}}$ is the weight matrix between input $f^i_{T_t}$ and memory cell c_t, and $\mathbf{W}_{\mathrm{ch}}$ is the weight matrix between hidden state h_{t-1} and memory cell c_t.

The output gate o_t can be seen as a filter that determines the information output from the memory cell c_t. The forward pass of the output gate can be expressed as

$$o_t = \sigma\left(\mathbf{W}_{\mathrm{oi}} f^i_{T_t} + \mathbf{W}_{\mathrm{oh}} h_{t-1} + b_o\right) \tag{13}$$

where $\mathbf{W}_{\mathrm{oi}}$ and $\mathbf{W}_{\mathrm{oh}}$ are the coefficient matrices of the output gate o_t and b_o is the bias vector. Then, the output of the current cell can be calculated by

$$h_t = o_t \tanh(c_t). \tag{14}$$

The change information can be learned by the present hidden state, and the outputs h^i_2 with temporal–spatial–spectral features are fed into the fully connected layers for classification. Hence, the probability estimation obtained by the fully connected layers can be seen as the final label prediction of the input sample. The final CD result $\mathbf{M}$ can be expressed as

$$\mathbf{M} = \mathrm{softmax}\left(\mathrm{Fc}\left(\mathrm{Concat}\left(\mathrm{Fc}(h^1_2), \mathrm{Fc}(h^2_2), \mathrm{Fc}(h^3_2)\right)\right)\right) \tag{15}$$

where $\mathrm{Fc}(\cdot)$ denotes the fully connected layers for dimension reduction and $\mathrm{softmax}(\cdot)$ indicates the softmax function.

C. Loss Function

Designing a reasonable loss function is a key step to obtain an ideal change map. Since the CD task can be regarded as a pixelwise binary classification, the commonly used cross-entropy loss L_{CE} is utilized during network training. The cross-entropy loss can measure the dissimilarity between the predicted value and the ground-truth label, so as to ensure that the change map is as close to ground truth as possible, which is defined as

$$L_{\mathrm{CE}} = -\frac{1}{n}\sum_{k=1}^{n}(y_k \log \hat{y}_k + (1 - y_k)\log(1 - \hat{y}_k)) \tag{16}$$

where n denotes the number of samples, y_k is the ground truth label of the given sample, and $\hat{y}_k$ is the predicted probability value.

Besides encouraging the loss to follow the minimization of the cross-entropy loss, the L_1 loss, which is the mean absolute error (MAE) distance, is added to constrain the reconstruction loss of the multilevel encoder–decoder subnetwork. By using such reconstruction loss to separately constrain two-stream encoder–decoder networks, the reconstructed output patch of the network is as close as possible to the input patch of the network so that the hidden layers in the middle can accurately and completely represent the multiscale spatial–spectral features of the input HSI. The reconstruction loss $L_{\mathrm{rec\_}t}$ for each patch can be expressed as

$$L_{\mathrm{rec\_}t} = -\frac{1}{n}\sum_{k=1}^{n}|\mathbf{I}_{T_t,k} - \widetilde{\mathbf{I}}_{T_t,k}| \tag{17}$$

where $\mathbf{I}_{T_t,k}$ and $\widetilde{\mathbf{I}}_{T_t,k}$ represent the kth sample of the input patch and the reconstructed patch, respectively.

In order to reduce the reconstruction error and the classification error simultaneously, a new loss function takes a weighted combination of the reconstruction loss of two-stream encoder–decoder networks and the cross-entropy loss. Hence, the overall loss function can be defined as

$$\begin{aligned} L_{\mathrm{overall}} = {} & \lambda_1 \underbrace{\left(-\frac{1}{n}\sum_{k=1}^{n}(y_k \log \hat{y}_k + (1 - y_k)\log(1 - \hat{y}_k))\right)}_{\text{cross-entropy loss}} \\ & + \lambda_2 \underbrace{\left(\frac{1}{n}\sum_{k=1}^{n}\left|\mathbf{I}_{T_1,k} - \widetilde{\mathbf{I}}_{T_1,k}\right|\right)}_{\text{reconstruction loss}} \\ & + \lambda_3 \underbrace{\left(\frac{1}{n}\sum_{k=1}^{n}\left|\mathbf{I}_{T_2,k} - \widetilde{\mathbf{I}}_{T_2,k}\right|\right)}_{\text{reconstruction loss}} \end{aligned} \tag{18}$$

where λ_1, λ_2, and λ_3 represent weighting coefficients.

III. Experiment Results and Analysis

A. Datasets

To demonstrate the effectiveness of the proposed model, we conducted experiments on three different datasets. Each dataset contains a ground-truth map and two HSIs taken at different times in the same area. The first two datasets are Yancheng and Jiangsu datasets and are acquired by the Earth Observing-1 (EO-1) Hyperion sensor [35], [36]. Another dataset is the BayArea dataset and was acquired by the Airborne Visible InfraRed Imaging Spectrometer (AVIRIS).

1) Hyperion Datasets: EO-1 was successfully launched by the National Aeronautics and Space Administration (NASA) on November 21, 2000, carrying three sensors: Hyperion sensor, advanced land imager (ALI) sensor, and atmospheric corrector (AC). The Yancheng and Jiangsu datasets are two datasets with a spectral resolution of 10 nm and a spatial resolution of 30 m taken by the Hyperion sensor within the spectral wavelength of 0.4–2.5 μm. Both of these two

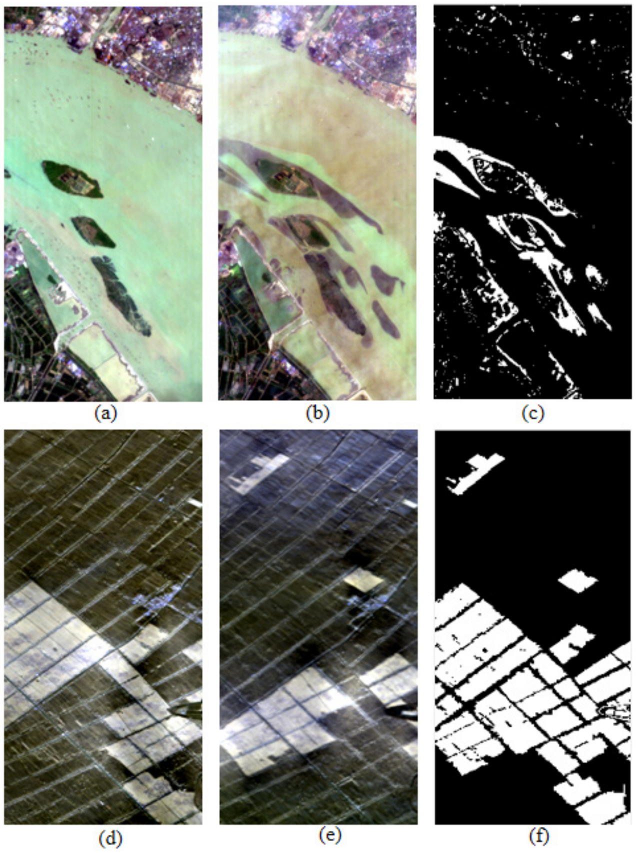

Fig. 2. Pseudo-color images of Yancheng and Jiangsu datasets acquired by EO-1 Hyperion sensor. (a) Jiangsu image acquired on May 3, 2013. (b) Jiangsu image acquired on December 31, 2013. (c) Ground truth of Jiangsu (the white area indicates change pixels and the black area indicates unchanged pixels). (d) Yancheng image acquired on May 3, 2006. (e) Yancheng image acquired on April 23, 2007. (f) Ground truth of Yancheng (the white area indicates change pixels and the black area indicates unchanged pixels).

datasets contain 242 spectral bands, in which some bands were polluted by noise in the course of collection, acquisition, and transmission. Therefore, in this article, we only choose the bands with a high signal-to-noise ratio for the experiment.

1) The first dataset[1] is Jiangsu dataset and was acquired by Hyperion sensor, as shown in Fig. 2(a) and (b), which mainly records the change of substance of a river in Jiangsu province, China. These two multitemporal images, which were taken on May 3, 2013 and December 31, 2013, are 463 $\times$ 241 pixels in size. In the experiments, 198 bands were used for CD task after discarding the noisy bands. Fig. 2(c) shows the ground truth of Jiangsu dataset, which contains 101 885 unchanged pixels and 9698 changed pixels.
2) The second dataset[2] is the Yancheng dataset and was acquired by Hyperion sensor, as shown in Fig. 2(d) and (e), which reflects the changes in farmland near the city of Yancheng, Jiangsu province, China. The two HSIs are captured on May 3, 2006 and April 23, 2007. After removing the noisy bands, each image has 154 bands with a size of 420 $\times$ 140 pixels. Fig. 2(f) shows the ground truth of Yancheng dataset, which contains 40 417 unchanged pixels and 18 383 changed pixels.

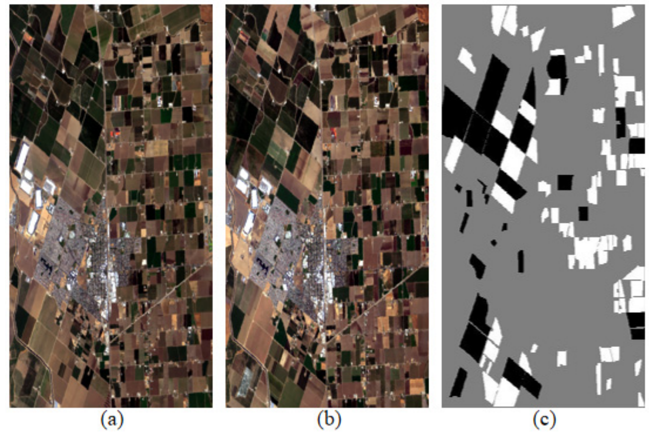

Fig. 3. Pseudo-color images of BayArea dataset acquired by AVIRIS sensor. (a) BayArea image acquired in 2013. (b) BayArea image acquired in 2015. (c) Ground truth of BayArea (the white area indicates change, the black area indicates unchanged, and the gray area is unmarked).

2) AVIRIS Dataset: AVIRIS, developed and maintained by NASA's Jet Propulsion Laboratory (JPL), is an advanced device for remote sensing of the Earth. It collects the spectral radiance information in the wavelength range of 0.4–2.5 μm and generates the HSIs with 224 continuous spectral bands. The HSIs are acquired by AVIRIS with the spatial resolution of 20 m and the spectral resolution of 10 nm.

1) BayArea dataset[3] is the third dataset and was acquired by AVIRIS, which mainly covers the city and the farmland of Patterson (California), USA, as shown in Fig. 3(a) and (b). The two HSIs were acquired in 2013 and 2015. These two images consist of 600 $\times$ 500 pixels, and 224 available bands were used for experiment. Fig. 3(c) shows the ground truth, in which the changed area (white) contains 39 270 pixels and the unchanged area (black) contains 34 211 pixels. The rest pixels (gray) of the ground truth are unmarked. In this dataset, only the tagged points are used for training and testing.

CD is regarded as a binary classification task, in which the training data can be divided into "changed" and "unchanged" classes. Since the proportions of these two classes are different in different datasets, we first calculate the number of changed and unchanged pixels in each dataset. Next, we randomly select 20% pixels from the changed and unchanged as training samples, and the remaining 80% pixels are used for testing.

B. Evaluation Measures

Generally, the change map is shown in the form of a binary image, in which the changed portions are denoted by white pixels and the unchanged parts are denoted by black pixels. In order to quantitatively evaluate the performance of the proposed method, two evaluation criteria are adopted in

[1]https://pan.baidu.com/s/14ht8k5H-8ObzHJS6msYZjQ
[2]https://rslab.ut.ac.ir/data
[3]https://citius.usc.es/investigacion/datasets/hyperspectral-change-detection-dataset

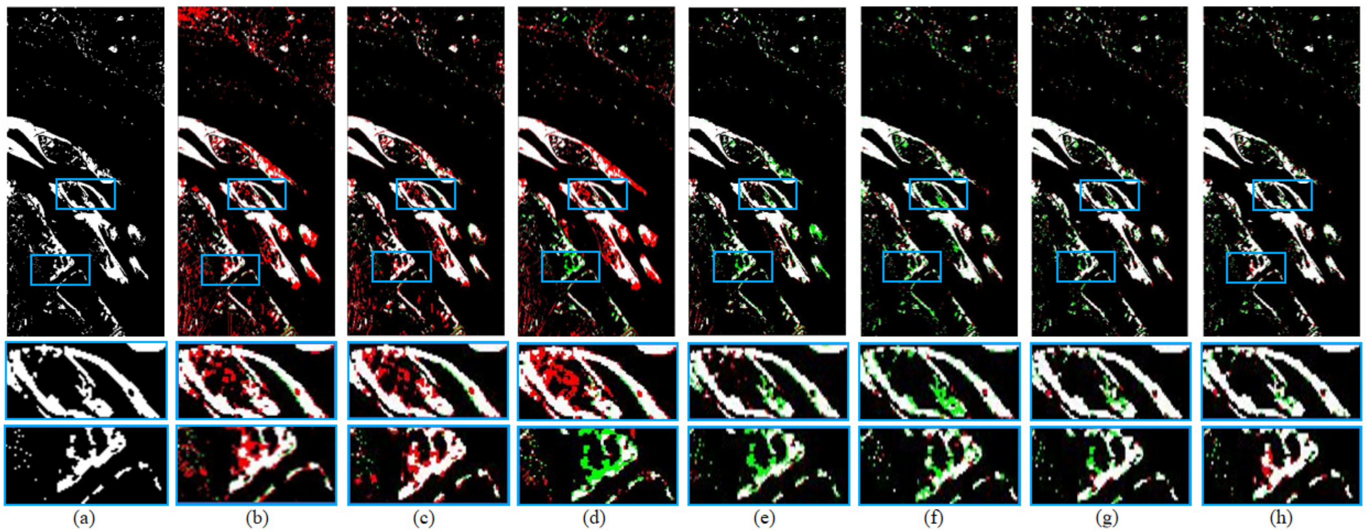

Fig. 4. Visualization results and two enlarged subregions of different methods on Jiangsu dataset. (a) Ground-truth map. (b) CVA. (c) PCA. (d) IR-MAD. (e) SVM. (f) ReCNN. (g) BCNNs. (h) ML-EDAN. Different colors are used for marking the pixels of TP, TN, FP, and FN, i.e., white for TP, black for TN, red for FP, and green for FN.

this article. In their calculation, true positive (TP) and true negative (TN) indicate the number of changed and unchanged pixels that are correctly detected, respectively. False positive (FP) denotes that the unchanged pixels in the ground truth map are erroneously detected as the changes in the change map, and false negative (FN) denotes that the changed pixels are misclassified as the unchanged class. Then, the overall accuracy (OA), which is adopted to access the OA of the CD method, can be calculated as

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}. \tag{19}$$

Kappa coefficient (KC) [37] is employed to evaluate the consistency between the CD results and the ground truth maps. Compared with OA, KC is more robust in evaluating classification accuracy. The higher the KC value is, the better the performance of the CD method will be. KC is calculated as

$$\mathrm{KC} = \frac{\mathrm{OA} - \mathrm{PRE}}{1 - \mathrm{PRE}} \tag{20}$$

$$\mathrm{PRE} = \frac{(\mathrm{TP} + \mathrm{FP})(\mathrm{TP} + \mathrm{FN})}{(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN})^2} + \frac{(\mathrm{FN} + \mathrm{TN})(\mathrm{FP} + \mathrm{TN})}{(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN})^2}. \tag{21}$$

C. Compared Methods and Experimental Setup

To verify the effectiveness of the proposed method, some representative approaches are selected as competing algorithms, including CVA [5], PCA [6], IR-MAD [8], support vector machines (SVMs) [38], ReCNN [25], and BCNNs [27]. CVA is a classic unsupervised approach, which uses the magnitude of the difference vectors to detect change area. PCA and IR-MAD are both transformation-based methods that are widely used in many CD tasks. SVM is selected as supervised classifiers and trained using radial basis function kernel in our experiments. ReCNN, a DL-based method, has achieved promising results by using LSTM architecture in the recurrent subnetwork. BCNNs, a recent DL-based method, are proposed in 2020 and utilize matrix cross multiplication to obtain the combined bilinear features. The feature map-based matrix cross multiplication can make full use of the features of the two images, thereby improving the accuracy of CD. These competing algorithms are chosen for their variety and effectiveness.

The proposed ML-EDAN is implemented on the PyTorch framework, and an NVIDIA GTX 3090 GPU is used for training and testing. The Adam optimizer [39] is employed for training and the initial learning rate is set to $1\mathrm{e}^{-4}$. Moreover, we train the proposed network for 200 epochs and the learning rate decays by a factor of 10 for every 150 epoch. The batch size for training is set to 16, and the parameter setups are the same for the three datasets.

It is generally considered that the spatial–spectral information in a local area is similar in HSIs. Two small patches cropped from the same position of two images taken at different times are used as the inputs of the network for the changed detection task. Since the spatial resolution of HSIs is relatively coarse, small patches can ensure the similarity of information in a local area while reducing the depth of the network. Therefore, in the experiments of this article, the patch size is set to 5.

In this article, considering the patches centered on the edge pixels of the HSIs, the edge of the image is first expanded in a way of mapping. Similar to the convolution operation, 5×5 patches are used as the sliding window for these two HSIs to generate samples, and labels are obtained from the corresponding position of ground truth, which is the center of the sliding window. Then, we randomly select 20% pixels from the changed and unchanged pixels as training samples.

D. Experimental Results

We have added the visual comparison of competing methods on three datasets, which is more suitable for human to perceive

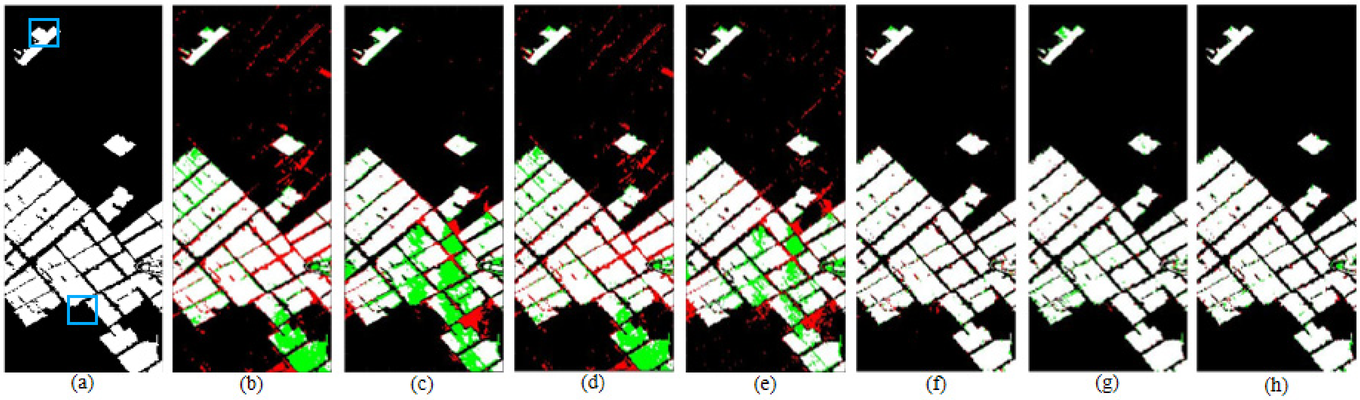

Fig. 5. Visualization results of different methods on the Yancheng dataset. (a) Ground-truth map. (b) CVA. (c) PCA. (d) IR-MAD. (e) SVM. (f) ReCNN. (g) BCNNs. (h) ML-EDAN. Different colors are used for marking the pixels of TP, TN, FP, and FN, i.e., white for TP, black for TN, red for FP, and green for FN.

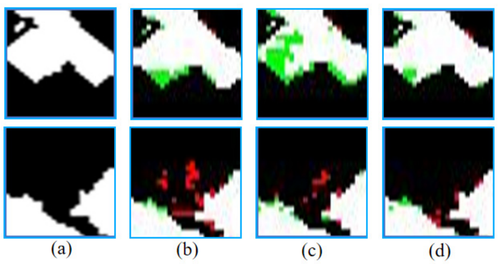

Fig. 6. Two enlarged subregions [shown in Fig. 5(a)] of CNN-based methods on the Yancheng dataset. (a) Ground-truth map. (b) ReCNN. (c) BCNNs. (d) ML-EDAN. Different colors are used for marking the pixels of TP, TN, FP, and FN, i.e., white for TP, black for TN, red for FP, and green for FN.

TABLE I

QUANTITATIVE ASSESSMENT OF ML-EDAN AND OTHER STATE-OF-THE-ART CD METHODS ON THE JIANGSU DATASET

Method	OA(%)	KAPPA	TP↑	TN↑	FP↓	FN↓
CVA	92.67	0.6575	9336	94073	7812	362
PCA	95.37	0.7538	9101	97314	4571	597
IR-MAD	93.83	0.6799	8557	96139	5746	1141
SVM	97.32	0.8244	7775	100823	1062	1923
ReCNN	97.14±0.10	0.8074±0.0094	7390	100895	990	2308
BCNNs	97.88±0.08	0.8627±0.0066	8199	101024	861	1499
ML-EDAN	**98.08±0.07**	**0.8771±0.0032**	8454	100988	897	1244

color. The visual results of the proposed and the competing algorithms on three datasets are shown in Figs. 4–8, where TP (white), TN (black), FP (red), and FN (green) are represented by different colors.

1) Experiments on the Jiangsu Dataset: For visual inspection, the change maps yielded by different methods on the Jiangsu dataset are shown in Fig. 4. It is challenging for all CD methods to identify the dispersed changed regions in the river. CVA cannot distinguish the unchanged regions very well, which results in a large number of red pixels on the change map, such as the enlarged subregions. Although the PCA approach reduces the number of red pixels, the performance of PCA is unsatisfactory for its poor ability to separate various changes. The result generated by IR-MAD can also recognize the major changes in the river and surrounding areas, but many isolated unchanged pixels shown in green pixels in Fig. 4(d) are misclassified into changed class, such as the area shown in the second enlarged subimage. Compared with CVA, PCA, and IR-MAD, SVM, ReCNN, and BCNNs achieve better performance, while many changed pixels marked by green are incorrectly labeled as unchanged pixels, as shown in Fig. 4(e)–(g). By contrast, the result of the proposed ML-EDAN method contains less green pixels, which shows that there is less phenomenon of detecting the changed regions as the unchanged regions. ML-EDAN has the least overall red and green pixels, indicating that the proposed method provides the best performance and can identify the irregular change well.

To quantitatively assess the performance of the proposed method, the quality analysis results based on OA and KAPPA are reported in Table I. The performance of CVA, PCA, and IR-MAD is not satisfactory. The results of ReCNN are superior to those of the unsupervised method, with higher OA and KAPPA values, since the training set is used to improve the performance of the methods. Due to the strong feature learning ability, the proposed method can learn the semantic difference between changed and unchanged regions so as to achieve higher KAPPA than SVM and BCNNs. As shown in Table I, the proposed method surpasses the competing methods in terms of all the objective metrics for CD task.

2) Experiments on the Yancheng Dataset: As shown in Fig. 5, the change map obtained from CVA is filled with many red points and green points, in which red points represent that unchanged pixels are incorrectly classified as the changed pixels and the green points indicate that the changed regions are failed to be recognized. This is mainly because the CVA based on the algebraic operation is sensitive to noise. PCA achieves better performance than CVA, while changes in some areas cannot be detected due to the loss of useful information caused by dimensionality reduction. Compared with CVA and PCA, IR-MAD has fewer unrecognized areas of change, as shown in Fig. 5(d). As a binary classification method, the SVM method achieves better performance with fewer red points and green points compared to the mentioned methods above, but the result is still unsatisfactory in visual. Although

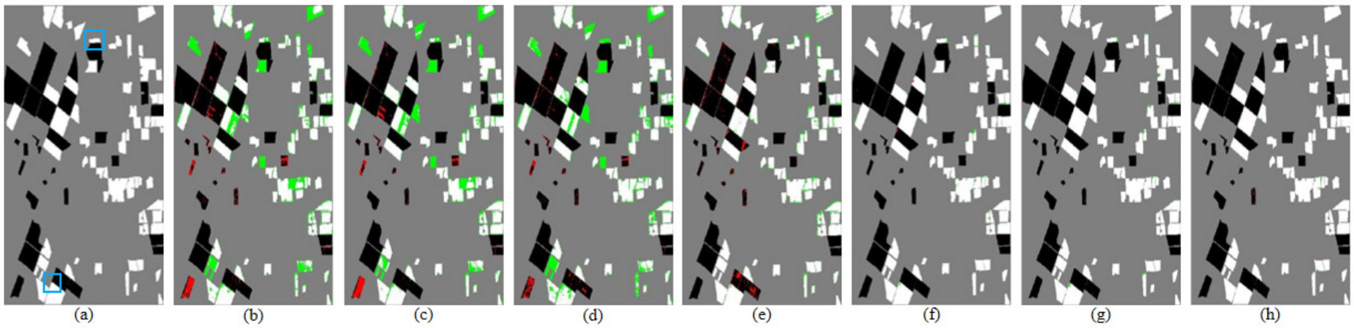

Fig. 7. Visualization results of different methods on the BayArea dataset. (a) Ground-truth map. (b) CVA. (c) PCA. (d) IR-MAD. (e) SVM. (f) ReCNN. (g) BCNNs. (h) ML-EDAN. Different colors are used for marking the pixels of TP, TN, FP, and FN, i.e., white for TP, black for TN, red for FP, and green for FN.

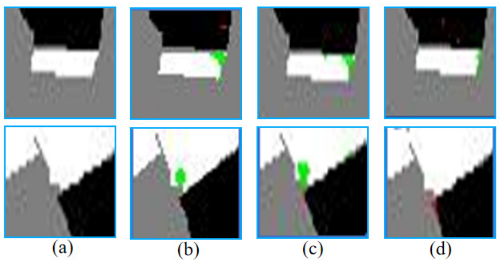

Fig. 8. Two enlarged subregions [shown in Fig. 7(a)] of CNN-based methods on the BayArea dataset. (a) Ground-truth map. (b) ReCNN. (c) BCNNs. (d) ML-EDAN. Different colors are used for marking the pixels of TP, TN, FP, and FN, i.e., white for TP, black for TN, red for FP, and green for FN.

TABLE II

QUANTITATIVE ASSESSMENT OF ML-EDAN AND OTHER STATE-OF-THE-ART CD METHODS ON THE YANCHENG DATASET

Method	OA(%)	KAPPA	TP↑	TN↑	FP↓	FN↓
CVA	87.49	0.6998	13689	37754	2663	4694
PCA	88.92	0.7322	13881	38402	2015	4502
IR-MAD	89.74	0.7632	15627	37142	3275	2756
SVM	91.15	0.7939	15735	37863	2554	2648
ReCNN	97.69±0.09	0.9461±0.0019	17813	39665	752	570
BCNNs	97.76±0.10	0.9488±0.0014	17479	40038	379	904
ML-EDAN	**98.08±0.11**	**0.9553±0.0026**	17758	39885	532	625

the CNN-based DL approaches ReCNN and BCNNs create a much better result compared to traditional methods, however, some errors still occurred at the edge of changed farmland, as shown in Fig. 5(f) and (g), respectively. As can be clearly seen, compared with ReCNN and BCNNs, ML-EDAN is able to effectively identify the changed regions in the corners of those independent regions, as shown in Fig. 5(h). For example, the subjective image of ML-EDAN has lesser green pixels than the subjective image of ReCNN and BCNNs in the top-left area. This is because the proposed method can reduce the probability of false detection and leak detection.

The quantitative analysis results of the Yancheng dataset are reported in Table II. Since the handcrafted features are weak in image representation, the traditional methods, including CVA, PCA, and IR-MAD, offer low OA and KAPPA. SVM performs better than the unsupervised methods in terms of both OA and KAPPA. The proposed method gets the highest detection accuracy on both OA and KCs. In particular, the proposed method achieves the best result with OA and KAPPA, higher than the DL-based ReCNN and BCNNs that are trained with the same samples as the proposed ML-EDAN. Thus, the result of the proposed method is proven to be superior to other competing methods in both qualitative and quantitative analyses.

TABLE III

QUANTITATIVE ASSESSMENT OF ML-EDAN AND OTHER STATE-OF-THE-ART CD METHODS ON THE BAYAREA DATASET

Method	OA(%)	KAPPA	TP↑	TN↑	FP↓	FN↓
CVA	84.06	0.6844	29510	32261	1950	9760
PCA	84.33	0.6897	29532	32434	1777	9738
IR-MAD	83.71	0.6782	28698	32814	1397	10572
SVM	81.55	0.6384	25819	34107	104	13451
ReCNN	99.15±0.10	0.9818±0.0017	34094	38720	550	117
BCNNs	99.24±0.13	0.9837±0.0020	34126	38829	441	85
ML-EDAN	**99.47±0.03**	**0.9894±0.0005**	34074	39034	236	137

3) Experiments on the BayArea Dataset: Fig. 7 shows the visualization results of all the competing approaches on the BayArea dataset. BayArea dataset consists mainly of cities and farmland. Due to the complexity of surface materials and the variety of crops, it is difficult for traditional methods to detect the changed areas, as shown in the green pixels in Fig. 7(b)–(e). In addition to SVM, other classical algorithms, CVA, PCA, and IR-MAD, also perform poorly for unchanged regions. As for Fig. 7(f)–(h), four statistical indicators, TP, TN, FP, and FN, are applied to quantitatively characterize the differences between three subjective images generated by ReCNN, BCNNs, and ML-EDAN.

Table III shows the quantitative assessment results of different methods on the BayArea dataset. As can be observed, all traditional methods yield relatively lower OA and KAPPA values than the DL methods, which is caused by the complexity and diversity of the substances and crops in the BayArea dataset. The performance of CVA, PCA, IR-MAD, and SVM is not satisfactory. Because DL-based ReCNN, BCNNs, and ML-EDAN algorithms can extract deep and complex features of ground objects that cannot be extracted by traditional algorithms, these DL methods can achieve OA of more than 99% and KAPPA of more than 0.98.

As shown in Tables I–III, the proposed ML-EDAN method achieves the best result with OA and KAPPA on three experimental datasets. BCNN obtains the second-best performance in terms of OA and KAPPA. For the TP, TN, FP, and FN indicators, the DL-based methods are better than the traditional CD methods. Compared with BCNNs, ML-EDAN yields larger TP values and smaller FN values on the Jiangsu and Yancheng datasets. These results demonstrate that for the Jiangsu and Yancheng datasets, ML-EDAN has a lower rate of missing inspection, and the false detection rate of BCNNs is lower. For the BayArea dataset, the TN and FP indicators of ML-EDAN are better, indicating that the false detection rate of ML-EDAN is lower on the BayArea dataset. Compared with BCNNs, although the TP, TN, FP, and FN of ML-EDAN are not all optimal, (TP + TN) of ML-EDAN is the largest and (FP + FN) of ML-EDAN is the smallest, which indicates that ML-EDAN can provide stable detection performance and the highest detection accuracy. As shown in Figs. 4, 5, and 7, the ReCNN, BCNNs, and ML-EDAN methods show acceptable and similar subjective results, especially the BCNNs and ML-EDAN methods. In order to further compare and analyze the performance of each method, Fig. 4 shows two enlarged subregions of each method of the Jiangsu dataset, and Figs. 6 and 8 show several enlarged subregions of ReCNN, BCNNs, and ML-EDAN methods of the Yancheng and BayArea datasets, respectively, where FP and FN are represented by red and green. It can be clearly seen that the CVA, PCA, IR-MAD, and SVM methods show quite a few red and green pixels. In several enlarged subregions, comparing RECNN and BCNNs, the ML-EDAN method shows the least red and green pixels, which indicates the excellent CD performance of the proposed method.

4) Computational Cost: We also tested the computational cost of all competing methods, and all experiments were implemented in MATLAB (R2015b) with Intel Core i5-7300HQ CPU@2.5 GHz CPU and NVIDIA GTX 3090 GPU. The computational costs of different methods on the Yancheng dataset are shown in Table IV, where the results are in seconds. It can be seen from Table IV that the traditional methods, CVA, PCA, and IR-MAD, have high efficiency, but the results generated by these algorithms are not satisfactory. Compared with traditional methods, SVM, ReCNN, BCNNs, and ML-EDAN are more time-consuming. ML-EDAN achieves the highest accuracy for CD compared to all competing algorithms. In the further study, we will consider to develop some more innovative model compression methods to reduce the computational cost of the method while ensuring the accuracy of CD task.

E. Ablation Study

1) Loss Function: The loss function of the proposed ML-EDAN is a weighted combination of the reconstruction loss and the cross-entropy loss [shown in (18)]. In (18), λ_1, λ_2, and λ_3 are weighting coefficients. In order to analyze the importance and effectiveness of the reconstruction loss and

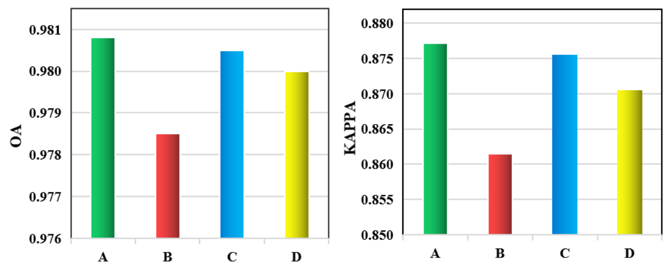

Fig. 9. Quantitative assessment results of ML-EDAN with different weighting coefficients λ_1, λ_2, and λ_3. (a) $\lambda_1 = 1$ and $\lambda_2 = \lambda_3 = 0.5$. (b) $\lambda_1 = 1$ and $\lambda_2 = \lambda_3 = 0$. (c) $\lambda_1 = \lambda_2 = \lambda_3 = 1$. (d) Dynamic λ_1, λ_2, and λ_3.

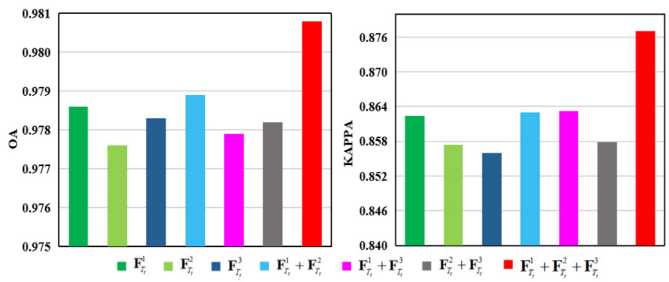

Fig. 10. Quantitative assessment results of ML-EDAN with different feature maps.

the cross-entropy loss and select the optimal values of the weighting coefficients, we trained the network with $\lambda_1 = 1$, $\lambda_2 = \lambda_3 = 0.5$, $\lambda_1 = \lambda_2 = \lambda_3 = 1$, dynamic learning λ_1, λ_2, λ_3, and $\lambda_1 = 1, \lambda_2 = \lambda_3 = 0$ on the Jiangsu dataset. Fig. 9 shows the quantitative assessment results of ML-EDAN with different weighting coefficients on the Jiangsu dataset. From Fig. 9, we can see that when $\lambda_1 = 1, \lambda_2 = \lambda_3 = 0$, the proposed network shows the worst performance. This illustrates that the proposed ML-EDAN with reconstruction loss achieves higher results in terms of OA and KAPPA indexes than ML-EDAN without reconstruction loss, which indicates the effectiveness of reconstruction loss. The results of fixed weighting coefficients $(\lambda_1 = 1, \lambda_2 = \lambda_3 = 0.5)$ are slightly higher than those of fixed weighting coefficients $(\lambda_1 = \lambda_2 = \lambda_3 = 1)$ and dynamic learning = coefficients. Therefore, in this article, the weighting coefficients λ_1, λ_2, and λ_3 are set to 1, 0.5, and 0.5, respectively.

2) Multilevel Fusion Strategy: In the proposed ML-EDAN method, we design a multilevel fusion strategy to make full use of spatial–spectral features at different scales. In order to analyze the effects of different combinations of $\mathbf{F}^1_{T_t}$, $\mathbf{F}^2_{T_t}$, and $\mathbf{F}^3_{T_t}$, we conduct the experiments on the Jiangsu dataset with increased feature maps, and the results of these experiments are shown in Fig. 10. As the number of feature maps increases, we can see a significant improvement in terms of KAPPA. This is because the multilevel spatial–spectral feature extraction subnetwork can extract features at different levels, among which low-level features can provide rich spatial details and high-level features can provide semantic information. The combination of features at different levels is more conducive to improving the accuracy of CD.

3) CIGA Module: In the proposed ML-EDAN method, we design a multilevel encoder–decoder subnetwork to extract hierarchical spatial–spectral features. The CIGA module is

TABLE IV
AVERAGE COMPETING TIME OF DIFFERENT ALGORITHMS ON THE YANCHENG DATASET

Method	CVA	PCA	IR-MAD	SVM		ReCNN		BCNNs		ML-EDAN	
				training	inference	training	inference	training	inference	training	inference
Time (s)	0.01	0.24	0.97	47.76	10.73	3392.40	3.53	4050.70	28.27	5342.72	33.63

TABLE V
QUANTITATIVE ASSESSMENT OF ML-EDAN WITH AND WITHOUT CIGA MODULE

Data set	Method	OA(%)	KAPPA
Jiangsu	w/o-CIGA	97.83	0.8604
	w-CIGA	**98.08**	**0.8771**
Yancheng	w/o-CIGA	97.74	0.9475
	w-CIGA	**98.03**	**0.9542**
BayArea	w/o-CIGA	99.01	0.9800
	w-CIGA	**99.49**	**0.9898**

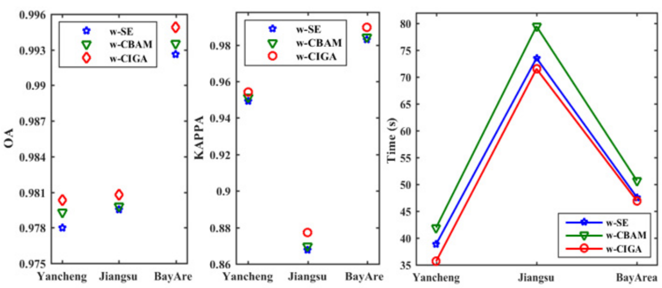

Fig. 11. Quantitative assessment results of ML-EDAN with different attention configurations.

embedded in the multilevel encoder–decoder subnetwork to yield more effective spatial–spectral features. To verify the effectiveness of CIGA, we compare the performance of ML-EDAN with and without the CIGA module, i.e., "w-CIGA" and "w/o-CIGA." The experimental results on three experimental datasets are shown in Table V, where the optimal values are highlighted in bold. It can be seen that for the three datasets, the results of "w-CIGA" are all better than those of "w/o-CIGA," especially the KAPPA index has obvious advantages, indicating that the CIGA module plays a great role in improving the detection performance of the network.

In order to further demonstrate the effect of the CIGA module, we compare the quantitative assessment results of ML-EDAN with different attention configurations. We compare our model embedded in the CIGA module (w-CIGA) with our model embedded in squeeze and excitation [40] block (w-SE) and convolutional block attention module (w-CBAM) [41]. The SE and CBAM blocks are two classic and effective attention mechanisms, and SE is a very simple module. We train the parameters of these three models so that they can obtain their best performance and provide the quantitative comparison of these three structures on three experimental datasets. The experimental results are shown in Fig. 11. It can be seen that w-CIGA achieves the best values on both OA and KAPPA, followed by w-CBAM, and finally w-SE. Through the comparison of the running times on three datasets, it can also be seen that w-CIGA has the lowest computational cost.

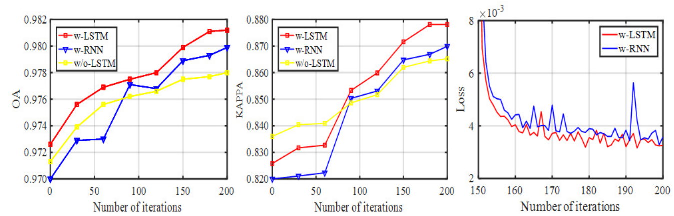

Fig. 12. Detection results of ML-EDAN w-LSTM and w/o-LSTM subnetwork.

We have also tested the performance of these models on a large number of multitemporal HS images and found that the proposed CIGA module can improve the performance and give the best results under the relatively low and acceptable calculation amount. This indicates that the CIGA module has the ability to highlight the information related to the CD task and extracts more spatial–spectral features to effectively improve the detection performance.

4) LSTM Subnetwork: The proposed method extracts the multilevel temporal–spatial–spectral features of bitemporal images by adopting the LSTM-based subnetwork. In order to study the effect of the LSTM-based subnetwork, we trained the proposed ML-EDAN method with RNN (w-RNN), LSTM (w-LSTM), and without LSTM (w/o-LSTM). Fig. 12 shows the detection results of different structures on the Jiangsu dataset. The w/o-LSTM structure is a pure CNN-based model, which obtains the multilevel spatial–spectral features of bitemporal images, without considering the temporal characteristics. The w-RNN and w-LSTM structures can extract multilevel temporal–spatial–spectral features of bitemporal images, which is beneficial to improve the performance of CD. The results shown in Fig. 12 verify this viewpoint, that is, when the compared structures all run to the optimal, the performance of w/o-LSTM is worse than that of w-RNN and w-LSTM. LSTM is a special RNN structure and is introduced to deal with the problems of gradient disappearance and gradient explosion during the RNN training process. As shown in Fig. 12, the results of w-LSTM are higher than those of w-RNN in terms of OA and KCs, and the proposed method ML-EDAN trained w-LSTM is more stable than trained w-RNN.

IV. CONCLUSION

In this article, we propose a novel network named ML-EDAN for hyperspectral image (HSI) CD, which integrates the architecture of encoder–decoder and LSTM. The multilevel encoder–decoder subnetwork is designed to exploit and fuse the hierarchical features, with CIGA module embedded to yield more effective spatial–spectral feature transfer. The LSTM-based subnetwork is further employed to learn global temporal features adaptively and jointly. Therefore, the proposed ML-EDAN is able to make full use of the hierarchical features and the temporal features from multitemporal HSIs. Moreover, the proposed network can be trained in an end-to-end manner. All these properties make ML-EDAN an effective approach to improve the performance of HS CD. The experiments on three real HSI datasets demonstrate that the proposed method achieves competitive performance and is superior to many state-of-the-art methods. Future works will focus on new approaches for training sample generation, such as unsupervised CD models, which select training data with high probabilities of being changed or unchanged in an appropriate way.

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